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**AMENDMENT No. 5
TO THE
PROCEDURES
FOR
AIR NAVIGATION SERVICES**

AERODROMES
(Doc 9981)

THIRD EDITION – 2020

INTERNATIONAL CIVIL AVIATION ORGANIZATION

Checklist of Amendments
to the PANS-Aerodromes (Doc 9981), Third Edition

	<i>Date of applicability</i>
Third Edition (incorporates Amendments 1 to 3)	5 November 2020 28 November 2024
Amendment No. 4 (approved by the President of the Council of ICAO on behalf of the Council on 19 June 2020)	4 November 2021
Amendment No. 5 (approved by the President of the Council of ICAO on behalf of the Council on 4 April 2025) Replacement pages viii, xiv, xv, xvii, xviii, I-4-App-23, I-4-Att A-5, II-4-2 to II-4-4, II-5-2, II-10-1 to II-10-11, II-10-App-1 to II-10-App-5 and II-10-Att-1	27 November 2025 21 November 2030



Transmittal note

Amendment No. 5

to the

Procedures for Air Navigation Services

AERODROMES

(Doc 9981)

1. Insert the following replacement pages in the PANS-Aerodromes (Third Edition) to incorporate Amendment No. 5, which becomes applicable on 27 November 2025 and 21 November 2030:

- | | |
|-------------------------------------|-------------------------------------|
| a) Page viii | — Contents |
| b) Pages xiv and xv | — Foreword |
| c) Pages xvii and xviii | — Acronyms and abbreviations |
| d) Page I-4-App-23 | — Part I, Appendix to Chapter 4 |
| e) Page I-4-Att A-5 | — Part I, Attachment A to Chapter 4 |
| f) Pages II-4-2 to II-4-4 | — Part II, Chapter 4 |
| g) Page II-5-2 | — Part II, Chapter 5 |
| h) Pages II-10-1 to II-10-11 | — Part II, Chapter 10 |
| i) Pages II-10-App-1 to II-10-App-5 | — Part II, Appendix to Chapter 10 |
| j) Page II-10-Att-1 | — Part II, Attachment to Chapter 10 |

2. Record the entry of this amendment on page iii.

Appendix 1 to Chapter 4. Reduced runway length operations	II-4-App 1-1
Appendix 2 to Chapter 4. Worksite control procedures.....	II-4-App 2-1
Attachment to Chapter 4. Worksite checklist.....	II-4-Att-1
Chapter 5. Foreign object debris (FOD) control	II-5-1
5.1 General	II-5-1
5.2 Objectives	II-5-1
5.3 Operational practices	II-5-2
Appendix 1 to Chapter 5. FOD training	II-5-App 1-1
Appendix 2 to Chapter 5. FOD prevention measures.....	II-5-App 2-1
Appendix 3 to Chapter 5. FOD detection, removal and evaluation.....	II-5-App 3-1
Chapter 6. Wildlife hazard management.....	II-6-1
6.1 General	II-6-1
6.2 Objectives	II-6-1
6.3 Operational practices	II-6-1
Appendix 1 to Chapter 6. Wildlife control training.....	II-6-App 1-1
Appendix 2 to Chapter 6. Wildlife incident reporting criteria	II-6-App 2-1
Attachment to Chapter 6. Land use on and around aerodromes	II-6-Att-1
Chapter 7. Apron safety	II-7-1
7.1 General	II-7-1
7.2 Objectives	II-7-1
7.3 Operational practices	II-7-2
Appendix to Chapter 7. Airside vehicle rules.....	II-7-App-1
Attachment to Chapter 7. Dissemination of information to apron users	II-7-Att-1
Chapter 8. Runway safety	II-8-1
8.1 General	II-8-1
8.2 Objectives	II-8-1
8.3 Operational practices	II-8-2
Appendix to Chapter 8. Runway safety team terms of reference and composition	II-8-App-1
Attachment A to Chapter 8. Best practice guide for “hot spots” identification, removal and promulgation	II-8-Att A-1
Attachment B to Chapter 8. Runway safety event causal factors.....	II-8-Att B-1

Attachment C to Chapter 8. Suspension or closure of runway operations	II-8-Att C-1
Chapter 9. Airside driver permit scheme and vehicle/equipment safety requirements.....	II-9-1
9.1 General	II-9-1
9.2 Objectives	II-9-1
9.3 Operational practices	II-9-2
Appendix 1 to Chapter 9. Framework for an airside vehicle driver training programme.....	II-9-App 1-1
Appendix 2 to Chapter 9. Airside roads and aprons — ADP	II-9-App 2-1
Appendix 3 to Chapter 9. Manoeuvring area permit.....	II-9-App 3-1
Appendix 4 to Chapter 9. Airside driver permit records	II-9-App 4-1
Chapter 10. Obstacle evaluation and control (<i>Applicable as of 21 November 2030</i>)	II-10-1
10.1 General	II-10-1
10.2 Objectives	II-10-2
10.3 Operational practices	II-10-3
Appendix to Chapter 10. Aeronautical study process (<i>Applicable as of 21 November 2030</i>)	II-10-App-1
Attachment to Chapter 10. Aeronautical study flowchart (<i>Applicable as of 21 November 2030</i>)	II-10-Att-1

6.12 Part II, Chapter 3 — Inspections of the movement area

6.12.1 This chapter contains provisions pertaining to inspections of the aerodrome movement areas (including runways, taxiways, aprons, and their associated aeronautical lighting systems, markings and signs), required to ensure the safe and efficient operation of aircraft.

6.12.2 The chapter outlines different types of inspections, their frequency, and relevant variables relating to the mix of aircraft operations, pavement materials and environmental conditions.

6.12.3 The appendix to Part II, Chapter 3 elaborates on the type and content of the inspections of the movement areas, and provides provisions on general procedures and documented reporting.

6.13 Part II, Chapter 4 — Work in progress (WIP)

6.13.1 This chapter details required procedures pertaining to the planning, coordination and safe execution of construction and other heavy works on the movement areas of aerodromes when aircraft operations continue in the works area.

6.13.2 The chapter lists possible precautions and considerations to be taken in order to ensure the safety of aerodrome operations, as well as the protection and safety of the worksite and workers.

6.13.3 The appendices and attachment to this chapter contain provisions on reduced runway length operations, and examples of supporting documents and checklists.

6.14 Part II, Chapter 5 — Foreign object debris (FOD) control

6.14.1 This chapter contains procedures and processes pertaining to the control and management of foreign object debris (FOD) at aerodromes, including the establishment of FOD control programmes to address the prevention, detection, removal and evaluation of FOD, as well as related training and awareness activities.

6.14.2 The appendices to this chapter detail provisions on FOD-related training, FOD sources, methods and techniques of FOD detection, and its removal, evaluation and reporting.

6.15 Part II, Chapter 6 — Wildlife hazard management

6.15.1 This chapter contains provisions relating to the reduction of the risk to aviation safety arising from wildlife, through the proactive management and control of wildlife at aerodromes and their vicinities.

6.15.2 The chapter details provisions on the establishment of a wildlife hazard management programme (WHMP) at aerodromes and WHMP components, including: expelling and deterring wildlife, reporting and recording wildlife incidents, habitat and land use management, and personnel training.

6.16 Part II, Chapter 7 — Apron safety

6.16.1 This chapter contains provisions on apron safety, including the identification of apron hazards and mitigation of risks, personnel awareness, the required coordination between the different parties involved in apron activities, and their responsibilities.

6.16.2 The appendix and attachment to this chapter deal with airside vehicle rules and the dissemination of information to apron users.

6.17 Part II, Chapter 8 — Runway safety

6.17.1 This chapter contains provisions pertaining to runway safety. The chapter deals with the prevention of runway incursions, excursions and confusion through different strategies. These include effective identification of hazards related to runway safety, risk mitigation, coordination and cooperation between different stakeholders, establishment of runway safety action plans and runway safety teams (RSTs), and the collection, sharing and exchange of safety information.

6.17.2 The appendix to Part II, Chapter 8 deals with RST composition and the terms of reference.

6.17.3 The attachments to Part II, Chapter 8 provide further detailed guidance on “hot spots”¹, common causes of runway safety events, and suspension or closure of runway operations.

6.18 Part II, Chapter 9 — Airside driver permit scheme and vehicle/equipment safety requirements

6.18.1 This chapter contains provisions on the establishment and implementation of an airside driver permit (ADP) scheme and safety requirements for vehicles and equipment operating at an aerodrome, with the objective of minimizing the risk of accidents causing injury to persons and damage to aircraft and property, arising from the use of vehicles in airside areas.

6.18.2 The appendices to Part II, Chapter 9 contain further detailed supporting provisions on the framework for an airside vehicle driver training programme, radiotelephony and record keeping.

6.19 Part II, Chapter 10 — Obstacle evaluation and control (Applicable as of 21 November 2030)

This chapter contains provisions pertaining to obstacle limitation surfaces and aeronautical study. It deals with topics on selection of required obstacle limitation surfaces, comprising obstacle free surfaces and obstacle evaluation surfaces, and how these surfaces can be adjusted to match the type of operations provided at an aerodrome. In addition, the chapter provides guidance on how to conduct aeronautical studies.

1. Definition from Annex 14 — *Aerodromes*, Volume I — *Aerodrome Design and Operations*:

“**Hot spot.** A location on an aerodrome movement area with a history or potential risk of collision or runway incursion, and where heightened attention by pilots/drivers is necessary.”

Table A. Amendments to the PANS-Aerodromes

<i>Amendment</i>	<i>Source(s)</i>	<i>Subject(s)</i>	<i>Approved Applicable</i>
1st Edition (2015)	PANS-Aerodromes Study Group (2009)	Procedures for Air Navigation Services — Aerodromes (PANS-Aerodromes).	20 October 2014 10 November 2016
1 2nd Edition (2016)	Friction Task Force (FTF) of the Aerodrome Design and Operations Panel (ADOP) (formerly the Aerodromes Panel (AP))	Amendment concerning the use of an enhanced global reporting format for assessing and reporting runway surface condition.	20 April 2016 10 November 2016 5 November 2020
2	Second meeting of the Aerodrome Design and Operations Panel (ADOP/2); Twelfth meeting of the Aeronautical Information Service (AIS) Aeronautical Information Management (AIM) Study Group (AIS-AIMSG/12)	Selection of engine for taxiway shoulder width; consequential amendment to changes to the SNOWTAM format as provided in the PANS-AIM (Doc 10066).	28 June 2018 8 November 2018 5 November 2020
3 3rd Edition (2020)	PANS-Aerodromes Study Group (PASG) in coordination with the Secretariat and the Aerodrome Design and Operations Panel (ADOP); and the third meeting of the Aerodrome Design and Operations Panel (ADOP/3)	Amendment concerning eight new chapters regarding day-to-day aerodrome operational management on training; inspections of the movement area; work in progress (WIP); foreign object debris (FOD) control; wildlife hazard management; apron safety; runway safety; and airside driver permit scheme and vehicle/equipment safety requirements; and a consequential amendment to the PANS-Aerodromes concerning aerodrome specifications. *Amendments related to pavement rating.	14 May 2020 5 November 2020 *28 November 2024
4	Eighth Meeting of the 220th Session of the ICAO Council	Postponement of the applicability date of Amendment 1 concerning the use of an enhanced global reporting format for assessing and reporting runway surface condition, excluding the editorial amendments concerning the structure of the document. Postponement of the applicability date of Amendment 2 consequential to changes to the SNOWTAM format as provided in the PANS-AIM (Doc 10066).	19 June 2020 4 November 2021
5	Fourth meeting of the Aerodrome Design and Operations Panel (ADOP/4)	a) visual aids; and b) obstacle limitation surfaces as of 21 November 2030.	4 April 2025 27 November 2025 21 November 2030

ACRONYMS AND ABBREVIATIONS

ACR	Aircraft classification rating
ADG [†]	Aeroplane design group
ADP	Airside driver permit
AGL	Above ground level
AIA	Accident investigation authority
AIP	Aeronautical information publication
AIS	Aeronautical information service
ANSP	Air navigation services provider
APAPI	Abbreviated precision approach path indicator
A-SMGCS	Advanced surface movement guidance and control system
ATIS	Automatic terminal information service
ATS	Air traffic service
AT-VASIS [†]	Abbreviated T visual approach slope indicator system
AVOL	Aerodrome visibility operational level
AVP	Airside vehicle permit
CAA	Civil aviation authority
CAD	Common agreement document
CDM	Collaborative decision-making
CFIT	Controlled flight into terrain
CNS	Communications, navigation and surveillance
FOD	Foreign object debris/damage
ft	Foot
GSE	Ground support equipment
IAIP	Integrated aeronautical information package
IAS	Indicated airspeed
IFR	Instrument flight rules
ILS	Instrument landing system
km	Kilometre
kt	Knot
LDA	Landing distance available
LNAV [†]	Lateral navigation
LVP	Low visibility procedures
MLS	Microwave landing system
NAVAID	Aid to air navigation
NDB	Non-directional radio beacon
NLA	New larger aeroplane
NPA	Non-precision approach procedure
OCA/H	Obstacle clearance altitude/height
OES [†]	Obstacle evaluation surface
OFS [†]	Obstacle free surface
OFZ	Obstacle free zone
OLS	Obstacle limitation surfaces
PAPI	Precision approach path indicator

[†] Applicable as of 21 November 2030.

PASG	PANS-Aerodromes Study Group (PASG)
PCR	Pavement classification rating
PRM	Precision runway monitor
QFU	Magnetic orientation of runway
RESA	Runway end safety area
RFF	Rescue and fire fighting
RNP [†]	Required navigation performance
RST	Runway safety team
RTF	Radiotelephony
RVR	Runway visual range
SARPs	Standards and Recommended Practices
SBAS	Satellite-based augmentation system
SDF [†]	Stepdown fix
SMGCS	Surface movement guidance and control system
SMS	Safety management system
SSP	State safety programme
TOD [†]	Terrain obstacle datasets
T-VASIS [†]	T visual approach slope indicator system
VASIS	Visual approach slope indicator system
VDGS	Visual docking guidance system
VFR	Visual flight rules
VNAV [†]	Vertical navigation
WGS-84	World Geodetic System — 1984
WHMP	Wildlife hazard management programme
WIP	Work in progress

[†] Applicable as of 21 November 2030.

10.3 Potential solutions can be developed by applying the following measures, alone or in combination with other measures. The following list is not in any particular order and is not exhaustive:

- a) adequate space on the pad to ensure a clear paved area around the aeroplane to facilitate the movement of de-icing/anti-icing vehicles;
- b) sufficient clearance between the pad and the adjacent manoeuvring areas taking the dimensions of aeroplanes into consideration;
- c) surface markings to ensure wing tip clearance of obstructions and other aeroplanes, especially if another aeroplane is also to be accommodated on the pad;
- d) the load bearing capacity of the existing structure;
- e) the requirement for greater quantities of de-icing/anti-icing agents;
- f) containment of excess run-off of de-icing/anti-icing agents;
- g) turning circle capabilities of specific aeroplanes;
- h) jet blast implications, especially in static breakaway and turns while exiting the facility, including the hazard to smaller aeroplanes nearby of possible degradation of agents; and
- i) revision of pad management procedures in terms of the positioning and exiting of aeroplanes versus smaller aeroplane types.

11. PAVEMENT DESIGN

Introduction

11.1 To facilitate flight planning, various aerodrome data are required to be published, such as data concerning the strength of pavements, which is one of the factors required to assess whether the aerodrome can be used by an aeroplane of a specific all-up mass.

Note.— The aircraft classification rating/pavement classification rating (ACR-PCR) method is used for reporting pavement strength. Requirements are given in Annex 14, Volume I, section 2.6, and Attachment A, section 20. Doc 9157, Part 3 — Pavements, contains guidance on reporting pavement strength using the ACR-PCR method.

11.2 The increased mass and/or gear load of the aeroplanes may require additional pavement support. Existing pavements and their maintenance will need to be evaluated for adequacy due to differences in wheel loading, tire pressure, and undercarriage design. Bridge, tunnel and culvert load-bearing capacities are a limiting factor, requiring some operational procedures.

Potential solutions

11.3 Potential solutions can be developed by applying the following measures, alone or in combination with other measures. The following list is not in any particular order and is not exhaustive:

- a) restrictions on aeroplanes with higher ACRs on specific taxiways, runway bridges or aprons; or
- b) adoption of adequate pavement maintenance programmes.

14. MAXIMUM AEROPLANE MASS

The maximum mass may have an impact on:

- a) the mass limitation on existing bridges, tunnels, culverts and other structures under runways and taxiways;
- b) disabled aeroplane removal;
- c) wake turbulence; and
- d) arresting systems when provided as an element of kinetic energy.

15. LANDING GEAR GEOMETRY, TIRE PRESSURE AND AIRCRAFT CLASSIFICATION RATING (ACR) VALUES

Landing gear geometry, tire pressure and ACR values may have an impact on the airfield pavement and associated shoulders.

16. ENGINE CHARACTERISTICS

16.1 The engine characteristics include engine geometry and engine airflow characteristics, which may affect aerodrome infrastructure as well as ground handling of the aeroplane and operations in adjacent areas which are likely to become affected by jet blast.

16.2 The engine geometry aspects are:

- a) the number of engines;
- b) the location of engines (span and length);
- c) the vertical clearance of engines; and
- d) the vertical and horizontal extent of possible jet blast or propeller wash.

16.3 The engine airflow characteristics are:

- a) idle, breakaway and take-off thrust exhaust velocities;
- b) thrust reverser fitment and flow patterns; and
- c) inlet suction effects at ground level.

16.4 The engine characteristics may be relevant for the following aerodrome infrastructure and operational aspects:

- a) runway shoulder width and composition (jet blast and ingestion issues during take-off and landing);
- b) shoulder width and composition of runway turn pads;
- c) taxiway shoulder width and composition (jet blast and ingestion issues during taxiing);
- d) bridge width (jet blast under the bridge);
- e) the dimensions and location of blast protection fences;
- f) the location and structural strength of signs;
- g) the characteristics of runway and taxiway edge lights;
- h) the separation between aeroplanes and adjacent ground service personnel, vehicles or passengers;
- i) snow removal procedures;
- j) the design of engine run-up areas and holding bays;
- k) the design and use of functional areas adjacent to the manoeuvring area;
- l) the design of air bridges; and
- m) the location of refuelling pits on the aircraft stand.

17. MAXIMUM PASSENGER- AND FUEL-CARRYING CAPACITY

Maximum passenger- and fuel-carrying capacity may have an impact on:

- a) terminal facilities;
- b) fuel storage and distribution;
- c) aerodrome emergency planning;
- d) aerodrome rescue and fire fighting; and
- e) air bridge loading configuration.

18. FLIGHT PERFORMANCE

Flight performance may have an impact on:

- a) runway width;

Chapter 4

WORK IN PROGRESS (WIP)

4.1 GENERAL

4.1.1 Periodically, construction and heavy maintenance activities are performed in movement areas. If aircraft operations are to continue around the site, or access is required through movement areas, there are a number of precautions to be taken to ensure the safety of aerodrome operations. These include the protection and safety of the worksite and workers.

4.1.2 Wherever major work affecting operational areas is planned, it is important that the aerodrome operator identifies and mitigates the risks generated by work in progress (WIP).

4.1.3 Part of effective safety management related to works lies in timely and comprehensive planning coordinated with all involved parties and relevant stakeholders.

4.1.4 Prior to their commencement, all measures necessary for the works to be undertaken safely, including timely notification of resulting operational changes, need to be communicated to all involved parties and relevant stakeholders.

4.2 OBJECTIVES

4.2.1 The aerodrome operator shall develop a process for managing the aerodrome operational safety during works on the movement area.

4.2.2 The process shall clearly establish and document the responsibilities and procedures for:

- a) the authorization of the works;
- b) the implementation of any proposed change to operational facilities;
- c) the date and time when the facilities will be withdrawn or changed from normal operations;
- d) the methods by which such changes will be promulgated;
- e) the oversight and control of the WIP; and
- f) the compliance with all relevant movement area safety rules.

4.3 OPERATIONAL PRACTICES

4.3.1 The process for managing the aerodrome operational safety during works should contain, at a minimum, the following elements:

- a) a works planning procedure;
- b) a procedure for a safety assessment of the planned changes to the operations or systems;
- c) a works authorization procedure;
- d) a procedure for the promulgation of information related to the works;
- e) a procedure for worksite establishment and return to aircraft operations; and
- f) a procedure for monitoring, oversight and control of the works.

Note 1.— The safety assessment methodology is included in Part I, Chapter 3 of this document.

Note 2.— The works authorization procedure identified in c) refers to local aerodrome operator approval for the commencement of works.

Note 3.— Annex 15 — Aeronautical Information Services includes appropriate methods on the promulgation of information.

Note 4.— The process for managing the aerodrome operational safety during works may be concentrated in one local procedure or be divided into several procedures, depending on local circumstances.

4.3.2 The works planning procedure should provide for the planning and coordination of works on the movement area so as to ensure that they are undertaken in a safe and compliant manner, while maintaining the operational safety, capacity, and efficiency of the aerodrome operations. During the planning process, representatives from aerodrome operations and other concerned stakeholders (such as ATS) should be involved at the earliest opportunity to consider the requirements for the efficient operations of the aerodrome during the proposed works.

4.3.3 A safety risk assessment of all planned works should be completed beforehand in order to ensure the hazards to the safe operation of aircraft have been identified by the aerodrome operator in coordination with interested parties, and appropriate mitigation measures introduced to keep risks to an acceptable level. Risk mitigation actions include, inter alia, the use of visual aids to denote restricted use area.

4.3.4 Processes, procedures, actions and decisions should be documented and made available to all relevant parties and stakeholders involved in the works or affected by any change in operations.

4.3.5 Before implementation, draft operational procedures, instructions or other information to be promulgated, should be discussed and coordinated with the directly affected stakeholders, and subjected to verification, thereby ensuring that their meaning is clear to potential users. Practical checks of proposed arrangements should be made by personnel having a comprehensive understanding of the operational implications of the works.

4.3.6 Where reduced runway length operations are required due to works, procedures shall be developed and implemented by the aerodrome operator, in accordance with provisions in Appendix 1 to this chapter.

4.3.7 Prior to the commencement of works, an authorization should be provided by the aerodrome operator to the party conducting the works. An authorization document should be used to ensure that specific permissions and conditions are recorded and agreed upon between the aerodrome operator and the relevant stakeholders. This document should also be communicated to the contractors, in order for them to be fully aware of what they can and cannot do.

4.3.8 The following actions should be taken when establishing the worksite, as well as throughout the duration of the works, when necessary:

- a) a safety risk assessment should be conducted to determine the need for visual aids to indicate temporary changes to the movement area;
- b) unserviceability markers are displayed when any portion of a taxiway, apron, or holding bay is unfit for the movement of aircraft, but it is still possible for aircraft to bypass the area safely;
- c) existing markings leading into a worksite should be masked or the route closed;
- d) unserviceability lights should be used and existing aeronautical ground lighting and signs leading into a worksite should be extinguished or masked on a movement area used at night or during low visibility;
- e) suitable site fencing should be installed to protect from jet blast, and FOD containment within the site should be enforced; and
- f) the perimeter of the worksite should be clearly marked and/or lit, particularly at night or during reduced visibility.

Note 1.— Unserviceability relates to areas temporarily not available for operational use.

Note 2.— A checklist for establishing worksites and returning them to operational use is included in the attachment to this chapter.

Note 3.— Unserviceability markers and lights are placed at sufficiently close intervals so as to clearly delineate the unserviceable area.

Note 4.— Measures for the situational awareness of pilots and drivers on the manoeuvring area need to take into consideration human factors that may lead to a runway incursion. Guidance material on Human Factors principles can be found in the Human Factors Training Manual (Doc 9683) and in the Manual on the Prevention of Runway Incursions (Doc 9870).

Note 5.— Temporary changes to the movement area may include, inter alia, reduction in the runway length, reduction in the maximum allowable wingspan, taxiway closure or any other closure to the movement area. Certain visual aids such as closed runway lighting could be used for a temporary period varying from a few hours to several weeks or longer, depending on the works in progress or other closure reasons.

4.3.9 Pre-startup and regular site meetings should be held to ensure that safety requirements are met and possible conflicts between the works and operations are resolved. Points to be considered include:

- a) safety awareness in relation to work on the movement area;
- b) workplace health and safety requirements;
- c) security requirements;
- d) protection of construction workers from aerodrome hazards, including jet blast and noise;

- e) procedure for quickly summoning emergency responders in case of a fire, spill, accident or similar event; and
- f) operational briefings on the interaction of the works with the aerodrome operations (e.g. runway(s) in use, likely visibility conditions, meteorological conditions, safety issues).

4.3.10 Procedures should be developed and implemented to monitor the safety of the aerodrome and aircraft operations in proximity of the works, such that timely corrective action is taken when necessary to assure their continued, safe operations. Additional procedures on worksite control are included in Appendix 2 to this chapter.

Chapter 5

FOREIGN OBJECT DEBRIS (FOD) CONTROL

5.1 GENERAL

5.1.1 The presence of foreign object debris (FOD) on the movement area may pose a significant hazard to the safety of aircraft operations. FOD has the potential to damage aircraft during critical phases of flight, which can lead to catastrophic loss of life and airframe, and increased maintenance and operating costs. FOD hazards can be reduced through the implementation of an FOD control programme, which would normally include FOD prevention, detection, removal and evaluation.

5.1.2 It is important that all personnel with access to the movement area understand their role in the prevention of FOD. FOD control is normally a module of the initial training given to personnel with access to the movement area.

5.1.3 It is necessary to have an established process to regularly clear the movement area of FOD. Removing FOD is the responsibility of everyone.

5.1.4 FOD may be controlled by ensuring that all personnel with movement area access, in particular inspection/maintenance personnel and ground handlers, are aware of situations which may potentially cause FOD.

5.2 OBJECTIVES

5.2.1 Aerodrome operators shall establish an FOD control programme commensurate with the assessed risks and appropriate to the local operating conditions. The programme shall consist of prevention, detection, removal and evaluation.

5.2.2 FOD prevention shall include awareness, training and education as well as measures for FOD prevention.

5.2.3 The detection of FOD shall include methods for the monitoring and inspection of the movement area.

5.2.4 Operational procedures and, where applicable, equipment, shall be provided for the removal, containment and disposal of FOD from the movement area.

5.2.5 Data and information about FOD shall be collected and analysed regularly to identify sources and trends.

5.3 OPERATIONAL PRACTICES

5.3.1 FOD prevention

5.3.1.1 Awareness

5.3.1.1.1 Aerodrome operators shall ensure that all personnel involved in aerodrome operations at the aerodrome are aware of the existence of the FOD control programme. Aerodrome personnel should be encouraged to identify potential FOD hazards, act to remove observed FOD, and propose solutions to mitigate related safety risks.

5.3.1.1.2 Aerodrome operators should designate an individual to manage the aerodrome's FOD control programme and clearly define the individual's responsibilities.

Note.— This designated individual can be a dedicated FOD manager or can have other duties (e.g. operations manager, safety manager).

5.3.1.1.3 The FOD control programme should be actively supported by the senior management of all organizations operating on the movement area.

5.3.1.1.4 Aerodromes operators should consider establishing an FOD committee to assist in the management of the FOD control programme, including the determination of potentially hazardous FOD situations and evaluation of collected FOD data. The committee may include stakeholders in a position to produce or remove FOD, such as ground handlers, aircraft operators, aerodrome operations and contractor representatives, etc.

Note.— The work of the FOD committee could be included in an already established safety-related committee (e.g. apron safety committee).

5.3.1.2 Training and education

The primary objectives of the FOD training programme are to increase employee awareness of the causes and effects of FOD damage and to promote active employee participation in eliminating FOD during the performance of daily work routines. A typical FOD training syllabus is included in Appendix 1 to this chapter.

5.3.1.3 FOD prevention measures

Active prevention measures to reduce the generation of FOD, commensurate with the identified safety risks, should be detailed in an aerodrome's FOD control programme. Details of these measures are included in Appendix 2 to this chapter.

5.3.2 FOD detection

5.3.2.1 Detecting FOD is an important activity at an aerodrome. This process involves not only the timely detection of any FOD, but also the identification of its potential source and location. Provisions on FOD detection processes, systems and equipment are included in Appendix 3 to this chapter.

5.3.2.2 Detection of FOD shall be included as part of the inspection regime.

Note.— See Part II, Chapter 3 — Inspections of the movement area, for the establishment of appropriate inspection procedures.

Chapter 10

OBSTACLE EVALUATION AND CONTROL

(Applicable as of 21 November 2030)

10.1 GENERAL

10.1.1 The airspace around aerodromes is critical to the safety and regularity of operations. These areas are often congested with aircraft that are performing landing and take-off operations. It is critical that this area be maintained free from obstacles that could adversely impact the safety and accessibility of intended aircraft operations at aerodromes. To meet this purpose, sets of obstacle limitation surfaces (OLS) were established to be introduced at aerodromes to prevent the aerodromes from being unusable due to uncontrolled growth of obstacles around aerodromes.

10.1.2 A new obstacle or the extension of an existing obstacle can adversely affect the safety and regularity of aircraft operations; it may result in increased operating minima, changes in air navigation service provider (ANSP) procedures, reduced capacity or operational restrictions on departures. It is therefore necessary to introduce OLS at and in the vicinity of aerodromes to effectively control the growth of obstacles and the possible turbulences that may be generated by these obstacles.

10.1.3 The existence of construction activities, buildings, structures, facilities, plantations, landfills or activities of any nature that interfere with the OLS may impose limitations on the effective utilization of an aerodrome or a portion of the airspace. It is therefore important for coordination to exist between Civil Aviation Authorities and aerodrome operators in managing the OLS.

10.1.4 The OLS defines the limits to which objects may project into the airspace and imposes restrictions on any public or private property, making them an effective method in controlling land use in the vicinity of aerodromes.

10.1.5 The OLS comprises two sets of surfaces: obstacle free surfaces (OFS) and obstacle evaluation surfaces (OES). The OFS and OES have distinct purposes and are applied based on the type of runway, aeroplane design group (ADG) and the flight procedures available for that runway.

10.1.6 The objectives and operational practices described in this chapter are intended to provide the information needed to apply the surfaces defined in Annex 14 – *Aerodromes*, Volume I – *Aerodrome Design and Operations*.

The surfaces

10.1.7 The OFS are surfaces that are applied within a defined airspace in the immediate vicinity of the aerodrome. The OFS are established to protect the existing and future operational capacity of the aerodrome by limiting obstacles. The OFS are intended to preserve accessibility of the aerodrome by containing standard operations (straight-in approaches) with a high level of probability. As such, they are to be kept free from obstacles except for existing obstacles and/or terrain which would have been assessed earlier.

10.1.8 The OES are surfaces that are applied in a defined airspace, in addition to the OFS, to be evaluated against obstacles. They are used in determining the acceptability of obstacles in ensuring safety and regularity of operations at the aerodrome. The penetration of the OES by terrain or obstacles are to be evaluated as they may adversely affect the safety or accessibility of the intended aircraft operations.

10.1.9 The dimensions of OFS and OES are determined based on ADG categorization. The ADG utilizes two criteria, which are related to the aeroplane performance characteristics and dimensions. The first criterion is based on the aircraft's indicated airspeed at threshold and the second criterion on the aeroplane wingspan.

10.1.10 In addition to existing operations, the OFS and OES selected for the aerodrome should safeguard future planned aircraft operations.

Aeronautical study

10.1.11 Objects penetrating the OFS and OES may adversely impact flight operations. The impact of these penetrations to safety and regularity of aircraft operations must be examined through the conduct of an aeronautical study.

10.1.12 An aeronautical study is a process of examining an aeronautical concern by assessing its impact on safety and regularity of aircraft operations and identifying, if need be, possible mitigation measures.

10.1.13 An aeronautical study may also be applied to examine the impact of obstacles at and in the vicinity of new airports or renovated and expanded airports at the master planning stage.

Note.— Further guidance on master planning can be found in the Airport Planning Manual (Doc 9184), Part I – Master Planning.

10.1.14 In addition to flight operations, the aeronautical study must consider and address other aeronautical concerns such as impact on communication facilities, navigational aids, aerodrome operations and air traffic control line of sight.

10.2 OBJECTIVES

10.2.1 States shall define the roles and responsibilities of stakeholder(s) and delegate them the appropriate authority in the identification of the OFS and required OES, the safeguarding of these surfaces and the surveillance of the areas within these surfaces.

Note.— Stakeholder(s) include aerodrome operators, local authorities, ANSPs, land use agencies, military organizations or any agency the State deems appropriate.

10.2.2 States shall establish a process for the identification of OFS and OES required to protect flight operations at an aerodrome.

10.2.3 Through the process, States or the appropriate authority shall:

- a) determine the OFS and OES applicable based on ADG category;
- b) vary the OFS, when required, to account for changes in the approach angle due to local conditions and aeroplane characteristics;

- c) determine the OES required to protect the operations at the aerodrome;
- d) vary the OES based on the operational requirements of the aerodrome;
- e) ensure the safeguarding of the OFS and OES; and
- f) identify obstacles and terrain data penetrating the OFS and OES.

Note.— The data collected are to be in accordance with terrain and obstacle datasets (TOD) provisions as stipulated in Annex 15 – Aeronautical Information Services and the Procedures for Air Navigation Services – Aeronautical Information Management (PANS-AIM, Doc 10066).

10.2.4 When instrument approaches are carried out on non-instrument runway(s), States or the appropriate authority shall review the dimensions of the approach and transitional OFS for such runway(s).

10.2.5 States shall establish a process for conducting the aeronautical study.

10.2.6 States shall define the party/parties responsible for conducting the aeronautical study.

10.2.7 The process to be established by States shall include the following details:

- a) identify stakeholders involved in the aeronautical study process;
- b) define situations where an aeronautical study is required;
- c) define the data required to produce a complete description of the aerodrome, obstacle and its environment;
- d) describe the methodology for the conduct of the impact assessment;
- e) identify appropriate mitigation measures to address any impact posed by an obstacle;
- f) define the acceptance criteria taking into consideration all studies and assessments submitted prior to approving the proposed obstacle; and
- g) define documentation and promulgation procedures of the aeronautical study.

Note.— An aeronautical study may be conducted anytime when, in the opinion of the State or appropriate authority, a proposed obstacle may adversely affect aircraft operations.

10.3 OPERATIONAL PRACTICES

10.3.1 Preventing the growth of obstacles

10.3.1.1 The primary responsibility to ensuring the aerodrome and its environment is and remains free from obstacles should be shared between the State and the aerodrome operator. The responsibilities and interactions among other stakeholders such as the aircraft operator, ANSP, local authority, etc., should be established and coordinated by the appropriate authority.

10.3.1.2 States are responsible for establishing a process for the selection of OLS required to protect the airspace against growth of obstacles and for evaluation of existing and potential obstacles and terrain. The process should include guidance on:

- a) identifying the OLS needed to support the existing and/or planned operations at the aerodrome;

Note.— For OLS selection, refer to 10.3.1.5.

- b) establishing the mechanisms to protect the surfaces and empower the appropriate authority to deny objects that may impact the safety and regularity of operations; and
- c) using OFS and OES for survey of the aerodrome and its surroundings and the collection of obstacle data.

10.3.1.3 The ADG consists of two criteria, indicated air speed at threshold and wingspan (see Table 10-1). In choosing the applicable ADG, the appropriate authority should consider the critical aeroplane operating at the aerodrome based on these two criteria (see Annex 14, Volume I, Chapter 1, 1.8.1 and 1.8.2).

Table 10-1. Aeroplane design group

<i>Aeroplane design group</i>	<i>Indicated airspeed at threshold</i>		<i>Wingspan</i>
I	Less than 169 km/h (91 kt)	and	Up to but not including 24 m
IIA	Less than 169 km/h (91 kt)	and	24 m up to but not including 36 m
IIB	169 km/h (91 kt) up to but not including 224 km/h (121 kt)	and	Up to but not including 36 m
IIC	224 km/h (121 kt) up to but not including 307 km/h (166 kt)	and	Up to but not including 36 m
III	Less than 307 km/h (166 kt)	and	36 m up to but not including 52 m
IV	Less than 307 km/h (166 kt)	and	52 m up to but not including 65 m
V	Less than 307 km/h (166 kt)	and	65 m up to but not including 80 m

10.3.1.4 In aerodromes with more than one runway, the appropriate authority can define different ADG for each runway and in areas where the surfaces overlap, the more stringent surface will apply.

Note.— Guidance on the application of ADG can be found in the Airport Services Manual (Doc 9137), Part 6 – Control of Obstacles.

10.3.1.5 The OLS to be adopted are dependent on:

- a) the type of runway – precision, non-precision and non-instrument;
- b) the ADG applicable for the runway; and
- c) existing and/or planned flight operations at an aerodrome and the corresponding OES(s) needed to protect the flight procedures. When selecting OES, the appropriate authority can adopt the OES defined in Annex 14, Volume I, Chapter 4, or design OES specific to the procedures conducted at the aerodrome.

10.3.1.6 States may delegate the responsibility for the safeguarding and surveying of areas within these surfaces to other appropriate authority.

10.3.1.7 The protection of these surfaces can be enhanced by incorporating them into the zoning laws. In lieu of zoning laws, the appropriate authority may also consider having the necessary coordination process with the land use agencies to ensure aviation requirements are factored into land use planning and airport master planning. The aviation requirements are not limited to OLS but may include other requirements relating to disturbance to communications, navigation and surveillance (CNS) facilities, glare or glint issues, wildlife management considerations and other potential hazards.

Note.— Further guidance can be found in the Airport Services Manual (Doc 9137), Part 6 – Control of Obstacles.

10.3.1.8 The surveying of aerodromes and their vicinity is key to ensuring safety of flight operations. The establishment of OFS and OES helps in ensuring the obstacle database remains updated. Objects that penetrate these surfaces are to be assessed and their information captured in the database.

10.3.1.9 A process shall be established to share the obstacle data with aeronautical information services (AIS).

10.3.1.10 The data collected, specifically the geographical coordinates of obstacles in TOD Area 2 and Area 3, should be measured and reported in degrees, minutes, seconds and tenths of seconds. In addition, the top elevation, obstacle type, marking and lighting (if any) of obstacles should be reported.

Note.— The TOD requirements including specifications on data quality can be found in Annex 15 and PANS-AIM, (Doc 10066), Table A1-6.

10.3.2 Adapting obstacle free surface and obstacle evaluation surface to operational needs

Adjusting obstacle free surfaces

10.3.2.1 The OFS are designed based on the nominal approach angle of 3.0°. There are aerodromes where the approach angle can either be lower or higher.

10.3.2.2 The slope of the approach surface is intended to adapt to approach operations that have an approach angle of 3.0° or higher. The slope of the approach surface may be raised if approach procedures with an approach angle of more than 3.0° are implemented. However, the approach angle should not be increased to enable the growth of obstacles.

10.3.2.3 Where the approach angle is less than 3.0°, the slope of the approach surface must be lowered to protect the operations.

10.3.2.4 In certain circumstances, due to alignment of visual slope indicator systems with instrument landings aids, the slope of the approach surface must be lowered, too, to align the surface with the requirements of the *Procedures for Air Navigation Services – Aircraft Operations* (Doc 8168).

Adjustments of the approach OFS

10.3.2.5 The approach OFS has been designed based on the approach angle of 3.0° and runway type. Any changes to these parameters will require changes to the dimensions of the OFS.

10.3.2.6 Changes to the approach angle will affect the slope and length of the approach OFS. As the OFS for instrument runway is designed to reach a height of 150 m (500 ft) or up to the obstacle clearance altitude/height (OCA/H), an increase or decrease in the slope will change the length of the surface.

Note.— Further guidance on changing the slope of OFS is given in the Airport Services Manual (Doc 9137), Part 6 – Control of Obstacles.

10.3.2.7 The slope of the approach OFS is associated with:

- a) the visual segment surface (see PANS-OPS (Doc 8168), Volume II, Part I, Section 4, Chapter 5, 5.4.6.2) which protects the visual phase of instrument approach procedures; and
- b) the obstacle protection surface of a precision approach path indicator (PAPI) for both instrument and non-instrument approach procedures.

10.3.2.8 To calculate the slope when there are changes to the approach angle, the following steps should be taken:

a) for runways served with instrument approaches:

- 1) subtract the standard angular margin from the promulgated approach angle. The standard angular margin equals the difference between the standard approach angle of 3.0° and the standard approach surface's gradient of 3.33 per cent which is approximately 1.91 per cent ($3.0^\circ - 3.33$ per cent);

Note.— Further guidance on the calculation of slope and their precalculated values is provided in Doc 9137, Part 6.

- 2) identify the slope associated with the obstacle protection surface. The harmonization between a 3.0° instrument landing system (ILS) and the PAPI will result in the obstacle protection surface of the PAPI to be at 3.22 per cent to align with the signal of the ILS (see Annex 14, Volume 1, Chapter 5, Figure 5-21 and Table 5-3). The slope of the approach surface shall be reduced accordingly; and

- 3) the lower slope between the two calculated in (i) and (ii) will be adopted for the approach OFS;

b) for runways intended for visual approach operations only:

- 1) when T visual approach slope indicator system (T-VASIS), abbreviated T visual approach slope indicator system (AT-VASIS), PAPI and abbreviated precision approach path indicator (APAPI) are provided, the obstacle protection surface will be used to determine the slope.

10.3.2.9 The length of the approach OFS is calculated to ensure the surface reaches a height of 150 m (500 ft). For an approach OFS slope of 4.1 per cent, the length is $500 \text{ ft} / 0.041 = 12\,195 \text{ ft}$ or 3.7 km. For instrument approaches, the length may need to be extended to reach the applicable OCA/H.

10.3.2.10 Where the final approach track is offset and intersects the extended runway centre line, the splay on the side closest to the final approach track should be increased by the offset angle. Where the final approach track is offset but does not intersect the extended runway centre line, the splay closest to the final approach track should be increased by an amount equal to the final approach course offset at 1 400 m from the runway threshold (see Figure 10-1).

Adjustment of the inner approach OFS

10.3.2.11 The inner approach OFS is part of the approach OFS. The surface will have the same slope as the approach OFS. Any adjustment to the slope of the approach OFS will bring about a corresponding change in the slope of the inner approach OFS. As the length of the inner approach extends to a height of 45 m, any variation to the slope will require adjustment to the length. The length of the inner approach can be calculated using the following formula:

Length of inner approach = 45 m/slope. For example, for an aerodrome with inner approach slope of 3.33 per cent, the required length is: $45 \text{ m} / 0.0333 = 1\,350 \text{ m}$.

Runway holding position and penetration of the approach OFS

10.3.2.12 The runway holding position can be established at the runway end or at the location prior to the landing threshold to reduce runway occupancy and increase runway capacity. This position is in accordance with the dimension stated in Table 3-2 of Annex 14, Volume 1. In any case, the following surfaces should not be penetrated:

- a) inner approach, inner transitional and balked landing surfaces (obstacle free surface): the inner approach, inner transitional and balked landing surfaces establish airspace for the protection of aeroplanes during balked landings and late go-arounds against fixed and mobile objects; and
- b) obstacle clearance surface: the OCS are introduced in PANS-OPS (Doc 8168) to deal with penetrations of the visual segment surface (VSS). Obstacles shall not require the pilot to destabilize the approach to avoid them. For this purpose, no obstacle shall penetrate an OCS.

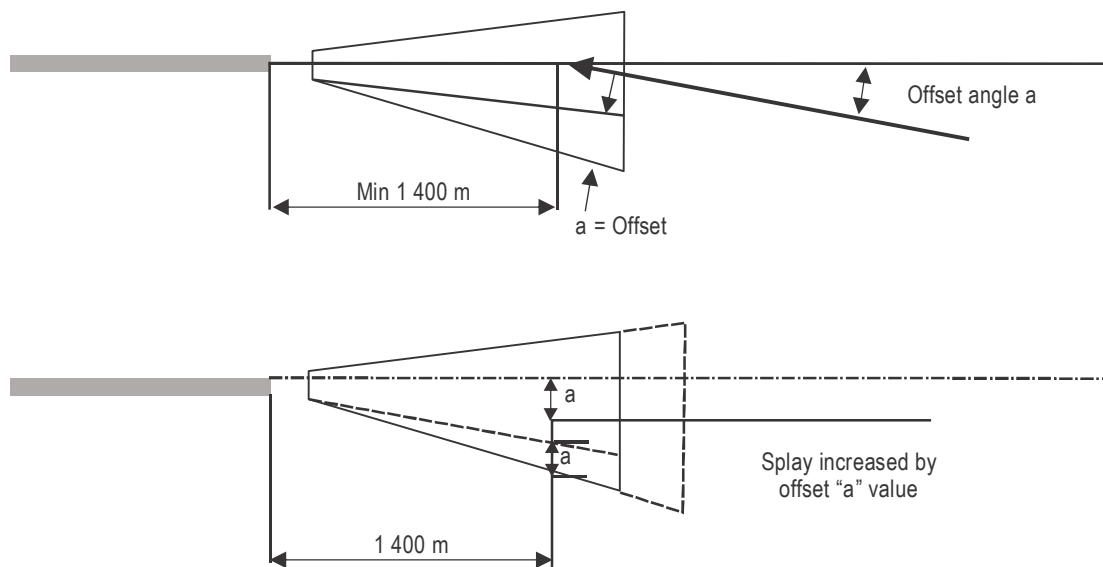


Figure 10-1. Plan view visual segment surface offset final approach with runway centre line crossing

Adjusting obstacle evaluation surfaces

10.3.2.13 The OES selected by the State or the appropriate authority should reflect the type of operations conducted and planned at the aerodrome. In selecting the OES, the appropriate authority should consider both existing and planned operations.

10.3.2.14 The appropriate authority can either:

- a) modify the OES to suit the existing and planned operational needs of the aerodrome;
- b) design an OES specific to the flight procedures conducted at the aerodrome; or
- c) declare an OES as an obstacle free surface.

Adjustment of the take-off climb surface

10.3.2.15 Aircraft during take-off may have different climb performances and therefore would be variably affected by an obstacle. At take-off, aircraft do not all lift off the runway from the same point nor do they follow the same climb rates. Hence, an object may be an obstacle for an aircraft with low performances on take-off but not to another aircraft with higher performance.

10.3.2.16 The slope of the take-off climb surface can be reviewed if the local conditions and the operational characteristics of aeroplanes of ADG IIC, III, IV and V operating at the runway support a slope higher than 2 per cent. Such review should be done in consultation with the aircraft operator as any increase to the slope may impact take-off performances and reduces payload.

10.3.2.17 A balanced approach to the selection of the gradient for the take-off climb surface is recommended. Local considerations need to be accounted for, as well as aircraft operator's requirements for efficient operations.

10.3.2.18 For typical aircraft operations with take-off masses not at maximum, the slope of the take-off climb surface may be increased to match the slope of the approach OES in the opposite runway direction when no significant limitations are to be expected, such as for narrow-body aircraft at low elevation aerodromes with longer runways.

10.3.2.19 In contrast, some aerodromes at higher elevations and/or in hot environments may require a slope of the take-off climb surface of less than 2.0 per cent.

Modifying horizontal OES

10.3.2.20 In situations where there is no circling or visual circuit, there is no need to establish the entire extent of horizontal OES. When a circling approach or visual circuit is not allowed on one side of the runway, the extent of horizontal OES may be reduced in order not to limit new constructions in that area.

10.3.2.21 Often, circling minima are not at the minimum due to the terrain and obstacle environment. In such cases, the height of the horizontal OES may be raised to match the obstacle clearance altitude (OCA) of the circling approach, considering the applicable minimum obstacle clearance requirements (see Figure 10-2).

Note 1.— A single or a combination of horizontal OES is to be adopted at aerodromes where there are circling operations.

Note 2.— It must be ensured that the modification of the horizontal OES will not impact the safety of visual procedures and aerodrome circuits, nor significantly affect the regularity of other flight procedures including instrument approach and departures procedures.

Declaring an OES as a surface that is free of obstacles

10.3.2.22 States or the appropriate authority may decide to declare an OES as a surface that should be free of obstacles. This may be done to protect the airspace in the vicinity of the aerodrome by restricting growth of obstacles that may undermine future aerodrome needs and flight operations. By not allowing penetration of the OES, it would simplify the process for approving obstacles, reducing the number of aeronautical studies required and reduce the likelihood of the flight procedures being amended.

10.3.2.23 When declaring an OES as a surface that should be maintained free of obstacles, amendments are necessary to the local zoning laws, aerodrome standards, etc.

Specific obstacle evaluation surfaces

10.3.2.24 In cases where existing and/or planned flight procedures cannot be safeguarded by the OES specified in Annex 14, Volume 1, Chapter 4, specific OES should be established to account for the local flight procedures.

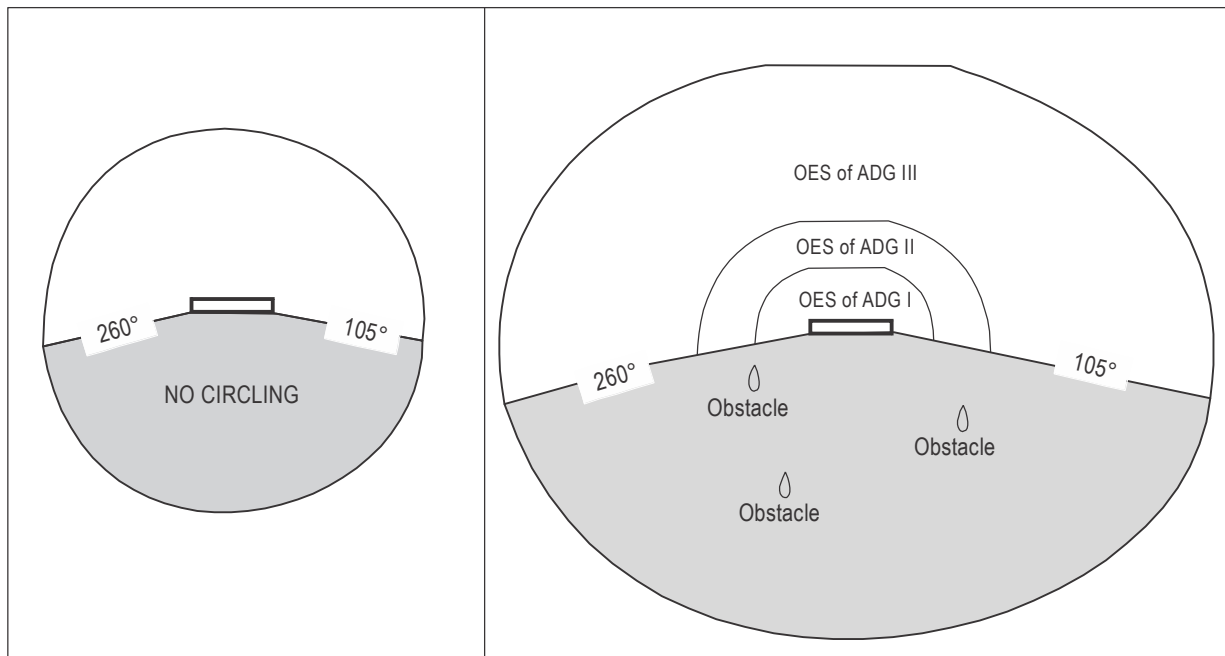


Figure 10-2. Modifying horizontal OES on a non-circling area

10.3.2.25 Adjustments of the OES or specific OES may be required for, including but not limited to:

- a) instrument approach procedures based on non-directional radio beacon (NDB) or radar;
- b) straight-in instrument approach procedures with low approach minima (obstacle clearance altitude/height (OCA/H));
- c) offset instrument approach procedures, or
- d) curved instrument approach procedures (Required Navigation Performance Authorization Required (RNP AR)).

10.3.2.26 While the horizontal surface(s) and surface for straight-in instrument approaches contained in Annex 14, Volume I, consider most common straight-in instrument approaches, the variety of all possible instrument approaches procedures cannot be addressed.

10.3.2.27 Precision approaches as well as straight-in approaches with significant offsets or curved approaches are not considered for the design of the surface for straight-in instrument approaches. Furthermore, low approach minima may not be fully addressed.

10.3.2.28 For non-precision approaches, approach minima of as low as 120 m (400 ft) may be achieved. With a minimum obstacle clearance (MOC) of 75 m (250 ft) for the final approach, an obstacle as high as the OES in the inner section (45 m or 150 ft) would result in approach minima of 120 m (400 ft). An obstacle as high as the OES in the outer section (60 m or 200 ft) would result in approach minima of 135 m (450 ft). An obstacle beyond the limits of the surfaces as high as 100 m (330 ft) would result in approach minima of 175 m (580 ft). In case the non-precision approach procedure supports stepdown fixes (SDFs), the approach minima of 175 m (580 ft) may be lowered to 135 m (450 ft) and to 120 m (400 ft).

10.3.2.29 For required navigation performance (RNP) approach procedures with lateral navigation (LNAV)/vertical navigation (VNAV) minima or localizer performance with vertical guidance (LPV) minima (satellite-based augmentation system (SBAS) approach procedure with vertical guidance (APV)-I), approach minima of approximately 85 m to 95 m (280 ft to 310 ft), depending on the aircraft speed category, or 100 m to 110 m (330 ft to 360 ft) may be achieved, depending on the actual approach procedure design as well as altitude and temperature effects, assuming obstacles as high as the surface for non-precision approaches.

10.3.2.30 Low approach minima are considered by means of the surface for precision approaches contained in Annex 14, Volume I. This surface, however, is based on the basic ILS surfaces and therefore considered precision approaches (using ILS, microwave landing system (MLS), GBAS, SBAS CAT I) only. For non-precision approach procedures (NPA) and APV with low approach minima (OCA/H), the dimensions of the surface for precision approaches in Annex 14, Volume I, is not sufficient and greater areas need to be considered.

10.3.2.31 The design of specific OES may be aligned with the protection areas specified for the design of instrument flight procedures. The related criteria are contained in PANS-OPS (Doc 8168) and the *Required Navigation Performance Authorization Required (RNP AR) Procedure Design Manual* (Doc 9905).

10.3.2.32 Furthermore, specific OES may be established to account for the local visual flight operations, including specified visual circuit patterns, VFR routes as well as VFR reporting points.

Note.— Guidance material for the protection of visual flight operations is contained in the Airport Services Manual (Doc 9137), Part 6 – Control of Obstacles.

10.3.3 Evaluating hazardous objects

10.3.3.1 Objects located beyond or below the OFS and OES may still be a concern to the safety and regularity of flight operations. These objects, due to their characteristics and purpose, may pose a hazard to air navigation. Such objects include wind turbines, chimneys, skeleton structures, transmission lines and power stations.

10.3.3.2 The orientation of buildings and their façades are known to cause turbulence, glare and disturbance to CNS equipment. The appropriate authority should work with land planning agencies to manage these hazards for buildings that are in the immediate vicinity of the aerodrome.

10.3.3.3 An aeronautical study should be considered when examining the impact on safety and regularity of flight operations due to these developments.

Note.— Details on the impact and references on possible assessment techniques to evaluate the risks posed by these objects can be found in the Airport Services Manual (Doc 9137), Part 6 – Control of Obstacles.

Appendix to Chapter 10

AERONAUTICAL STUDY PROCESS

(Applicable as of 21 November 2030)

1. Aeronautical study

1.1 To investigate an aeronautical concern arising from the introduction of an obstacle, an aeronautical study is used to evaluate the impact of the obstacle on flight operations and identify possible measures that can mitigate these concerns.

1.2 The process of conducting an aeronautical study should include:

- a) data gathering;
- b) stakeholder engagement and impact assessment;
- c) identifying mitigations (if applicable); and
- d) acceptance or rejection.

Note.— A flowchart showing the aeronautical study process is included as an Attachment to this Appendix.

2. Data gathering

2.1 Information collected should include but not be limited to:

- a) national regulations and procedures (zoning laws, land use policies);
- b) use of the aerodrome (day/night, IFR/VFR, public/private use, certified/uncertified);
- c) dimensions of approach lighting system;
- d) existing and future runway and taxiway characteristics;
- e) mix of aircraft operating at the aerodrome;
- f) number of regular flights (commercial air transport);
- g) procedures specific to the aerodrome;
- h) flight procedures (existing and future);
- i) contingency procedures;
- j) details on existing obstacles and mitigations;

- k) existing and planned visual navigation aids, electronic navigation aids, surveillance aids, communication aids, etc.; and
- l) airspace structure and details of nearby aerodromes.

Note.— For non-flight operations related aeronautical studies, other information may be required.

2.2 Details of the proposed obstacle or terrain should include but not be limited to:

- a) location;
- b) obstacle evaluation;
- c) classification (such as building, crane, tree(s), antenna(s), power lines);
- d) dimensions as well as height, top elevation including ground elevation;
- e) frangibility;
- f) permanence (permanent or temporary);
- g) presence (fixed or mobile);
- h) visibility (such as marking and lighting); and
- i) material (as it may interfere with electronic signals).

2.3 Aerodrome details should include (this list is not exhaustive and may include others):

- a) ICAO code;
- b) CNS facilities;
- c) procedures specific to the aerodrome;
- d) use of aerodrome (IFR/VFR day/low visibility operations)
- e) visual aids;
- f) type of aircraft;
- g) runway characteristics (number of runways, dimensions, type);
- h) airspace;
- i) nearby aerodromes; and
- j) aerodrome noise zones.

2.4 Affected surfaces should include:

- a) surfaces penetrated; and
- b) extent of penetration.

2.5 Flight operations related to instrument flight procedures should include (this list is not exhaustive and may include others):

- a) instrument approach procedures (initial, intermediate, final and missed approaches as well as visual segment and circling);
- b) standard instrument departures (SIDs) (standard instrument departure routes and/or omni-directional departures);
- c) standard instrument arrival (STAR) routes;
- d) minimum sector altitudes (MSA) and terminal arrival altitudes (TAA);
- e) holding patterns;
- f) ATC surveillance minimum altitudes; and
- g) en route ATS routes.

2.6 Flight operations should include (visual flight procedures and visual approaches):

- a) visual departures;
- b) traffic patterns (standard circuit patterns or other published patterns, including patterns for entry and exit to/from circuit patterns); and
- c) VFR routes and VFR reporting points.

3. Stakeholder engagement and impact assessment

3.1 The proponent of the aeronautical study can either be an agency of the State, aerodrome operators or by any appropriate authority. In conducting the study, it is critical to have the participation of all relevant stakeholders. The stakeholders should include but are not limited to:

- a) the Civil Aviation Authority;
- b) proponent of the proposed construction or development;
- c) the aerodrome operator;
- d) air navigation services providers;
- e) flight procedure designers;
- f) aircraft operators (commercial aviation and the military); and
- g) other appropriate authorities (such as land use planner, military, etc.).

Note 1.— The level of stakeholders' involvement may vary depending on the needs of the study.

Note 2.— The design of procedures in accordance with PANS-OPS (Doc 8168) criteria assumes normal operations. It is the responsibility of the operator to provide contingency procedures for abnormal and emergency operations. It might be necessary to involve aircraft operators in the aeronautical study process.

3.2 There could be a situation where more than one OES is penetrated. Due to different purposes each OES serves, every surface needs to be evaluated individually to assess its impact.

Note.— Explanations on how impact assessments are conducted can be found in the Airport Services Manual (Doc 9137), Part 6 – Control of Obstacles.

4. Identifying mitigations

Note.— Identifying the mitigating measure and assessing their impact on stakeholders is a key task in the aeronautical study process. Different measures are required to address OFS and OES penetrations. Certain mitigation measures, while acceptable, may not be desirable due to the penalty they impose on aerodrome operations.

4.1 The mitigation measure identified should consider the obstacle limitation surface(s) that is (are) being penetrated.

4.2 Obstacles penetrating the OFS will impose an adverse effect on the safety of flight operations and should be removed. Where removal is not practicable, the penetrations are to be mitigated by means of limiting operations at the aerodrome, such as designating the runway for departure only or allowing only visual approaches to the runway, before considering other measures. Other measures include, but are not limited to:

- a) displacing the threshold; and/or
- b) raising the glide slope and PAPI angle.

4.3 Obstacles penetrating the OES may have an adverse impact on the safety or accessibility of intended aircraft operations. Mitigation measures may include, but are not limited to:

- a) adjusting flight procedures (such as increased minimum obstacle clearance altitudes, increased minimum climb gradients, change of routing);
- b) including the obstacles in the relevant ICAO charts;
- c) promulgating safety information to the users of the aerodrome (particularly concerning the change);
- d) increasing approach minima (OCA/H);
- e) increasing minimum climb gradient for departures (procedure design gradient (PDG));
- f) increasing descent gradients;
- g) increasing flight altitude;
- h) revising traffic and circuit patterns;
- i) revising VFR routes and/or reporting points; and

- j) adjusting and including marking and lighting.

4.4 Upon completion of the aeronautical study, it may be necessary to conduct a safety risk assessment to quantify the risk probability and severity and the acceptability of any proposed mitigation measures in reducing the risk to an acceptable level.

Note.— Provisions on safety risk assessment are stipulated in Chapter 3 of this document and in the Safety Management Manual (Doc 9859).

5. Acceptance

5.1 The study may result in one of the following outcomes, where the obstacle is:

- a) acceptable, because the risk is already mitigated;
- b) acceptable, only if the risk is mitigated; or
- c) not acceptable, because the risk cannot be mitigated.

5.2 The State or the appropriate authority establishes the approval and acceptance criteria to be used in evaluating the study. The study is to be submitted by the aerodrome operator or an appropriate authority prior to development or installation of the proposed obstacle.

5.3 The State or the appropriate authority should analyse the aeronautical study and verify that:

- a) appropriate coordination has been performed between the stakeholders;
- b) aeronautical concerns have been properly investigated and assessed, based on current and future flight operations and procedures;
- c) proposed mitigation measures adequately address the impact posed by the obstacle; and
- d) the subsequent safety risk assessment, when required, has been conducted in accordance with Chapter 3 of this document and meets the acceptability criteria specified in 3.5.1.

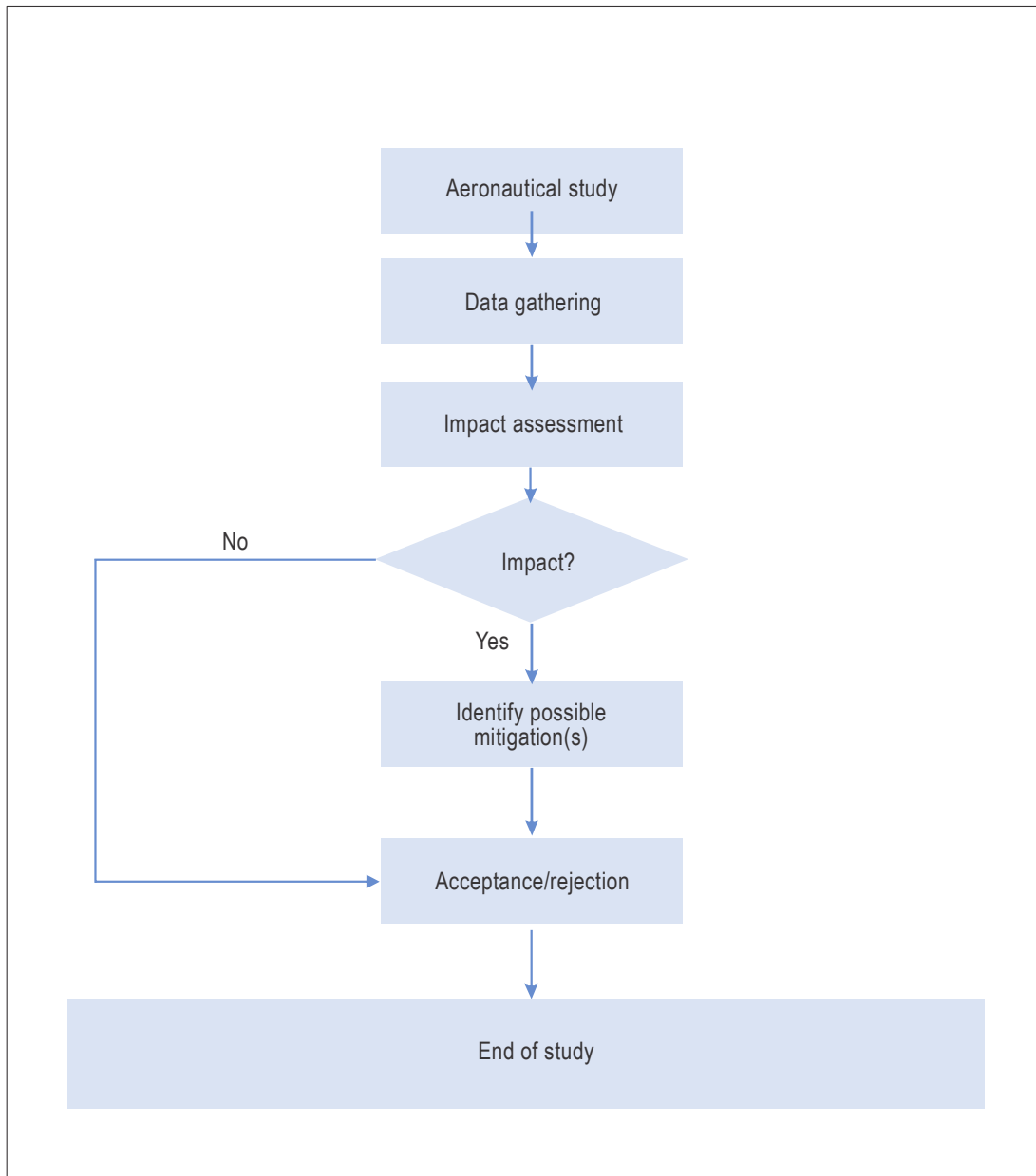
Note.— It is preferable to have relevant experts from the State, as required, on the team that conducts the aeronautical study. On completion of the analysis of the aeronautical study, the State or the appropriate authority:

- a) gives formal approval or acceptance of the aeronautical study to the aerodrome operator or an appropriate authority. If in the study some impacts to flight operations have been underestimated or have not been identified, coordinate with the aerodrome operator or an appropriate authority to reach an agreement on safety acceptance; or*
- b) if no agreement can be reached, rejects the proposal for possible resubmission by the aerodrome operator or an appropriate authority; or*
- c) may choose to impose conditional measures to ensure safety and regularity.*

Attachment to Chapter 10

AERONAUTICAL STUDY FLOWCHART

(Applicable as of 21 November 2030)



— END —

